Flexible Long-range Synchronization of fast brain rhythms through chaos

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(Dated: March 18, 2026)

Gamma rhythm (30-100 Hz) local field potentials recorded from distant brain regions show a strong coherence. The existence of such long-range synchronization poses significant challenges to the current belief that gamma rhythms are primarily involved in local computation and function. Notably, the exact mechanisms of such synchronization remain unclear. Here, we uncover a new mechanism by which chaotic, slow rhythms (theta or alpha) induce gamma rhythms and coordinate their timing. The chaotic activity of two coupled slow rhythms acts as inputs that generate gamma rhythms in two distant brain regions, providing the randomness that characterizes them. Surprisingly, they also enable the coordination required for long-range synchronization via a Phase-phase coupling mechanism. The long-range phase synchronization exhibits regimes of symmetric and asymmetric phase offsets, indicating distinct routes of bidirectional communication. In each regime, the transition between the two phase offsets occurs dynamically and is governed by the chaotic dynamics of the slow rhythms. Our mechanism is robust and flexible, suggesting that gamma rhythms may be involved in global computation.

*Introduction.*Gamma rhythms (30-100 Hz) recorded in brain areas such as the hippocampus [1] and the visual cortex are broadband with a fast temporal decorrelation [2]. They are usually coupled to slower theta rhythms (4-12 Hz). While gamma rhythms are implicated in local communication between nearby areas oscillating in the same frequency band, like between V1 and V2 [3] and between V1 and V4 [4], they are not expected to foster communication between distant areas [5]. However, phase synchronization at gamma frequency between the Left and Right primary visual cortex hemispheres of anesthetized cats has been observed [6]. Similarly, coherence has been observed between the Local field potentials of hippocampal CA1 pyramidal layers in the left and right hemispheres of a cat during running and sleep [7]. This suggests that local gamma rhythms can facilitate long-range communication, raising the question of the underlying mechanism of this ability.

A tantalizing proposal for this general phenomenon involves within-region cross-frequency coupling between slow and fast rhythms, combined with long-range phase synchronization between slow rhythms across regions [8], for which we provide a dynamical mechanism. A slow theta rhythm is seen as a coherent self-sustained oscillation [1] that can mediate synchronization at larger scales than a gamma rhythm [5]. By modulating the gamma rhythms to which they are locally coupled via phase-amplitude or phase-phase cross-frequency coupling [8, 9], they can coordinate these rhythms to achieve distant synchronization. Dynamical analysis reveals that, paradoxically, chaos is necessary for this proposal to unfold, as we now explain.

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The model. We consider the system of coupled Wilson-Cowan networks [10] described by the first-order coupled differential equations

$$\dot{E}_k^r(t) = -\alpha_E E_k^r + (1 - E_k^r)\beta_E f(S_{E_k}^r) \quad (1)$$

$$\dot{I}_k^r(t) = -\alpha_{I_k} I_k^r + (1 - I_k^r)\beta_I f(S_{I_k}^r) \quad (2)$$

$$S_{E_k}^r = W_{EE_k} E_k^r - W_{EI} I_k^r + h_E + c_{EE}^k E_f^r + c_L^k E_s^r \quad (3)$$

$$S_{I_k}^r = W_{IE} E_k^r - W_{II} I_k^r + h_I \quad (4)$$

with $r = 1, 2$ and $k = s, f$. The notations s and f represent the slow and fast networks, respectively, and the superscripts $r = 1, 2$ refer to the first and second regions, respectively. For the sake of clarity, a population is composed of either excitatory or inhibitory neurons represented in red and blue in Fig.1-**A**. At the same time, a "network" refers to a coupled population of excitatory and inhibitory neurons. A fast network generates gamma rhythms, while a slow network generates theta rhythms. Fast and slow unidirectional, coupled networks constitute a "region." However, we will continue to refer to slow rhythms as those generated by the slow networks and to fast rhythms as those generated by the fast networks, even when they may contain both slow and fast components due to coupling with the slow and fast networks. The two regions in our model are identical, and the slow and fast networks differ only by the values of the inhibitory decay rate $\alpha_{I_s} < \alpha_{I_f}$ and the recurrent excitatory coupling $W_{EE_s} > W_{EE_f}$. With this choice, the slow network generates self-sustained (limit cycle) oscillations when all the local c_{EE}^k and long-range c_L^k couplings are set to zero, and the fast network spirals towards a stable fixed point. When the local couplings are considered, each region generates coupled fast and slow rhythms. We are interested in the ability of the fast brain rhythms to establish long-range synchronization

and, therefore, communication. The long-range coupling c_L^k is symmetric and targets only excitatory neurons ($c_L^f = 0$ and $c_L^s = c_L$) of the slow networks of each region. In addition, the connectivity between slow and fast networks in a region is unidirectional from the slow to the fast network ($c_{EE}^f = 0$ and $c_{EE}^s = c_{EE}$) (see Fig.1-**A**). The model built with these parameters exhibits cross-frequency coupling between local fast-slow networks for different values of the long-range excitatory coupling. This network architecture has recently attracted attention as a realistic model of brain connectivity [8, 11].

The dynamics of the model. We are interested in long-range synchronization between the two fast brain rhythms as a substrate for communication [12]. The parameters of each fast network are chosen so that the corresponding fixed point is stable and the dynamics decay when the fast network is not receiving input from the slow networks. Therefore, we only investigate the dynamics of the coupled slow network as the long-range coupling c_L varies. We simulate the system of slow/fast networks (Fig.1-**A**) for different values of c_L (Fig.1-**B-C**) and focus on several quantities including the slow rhythms and their peaks (Fig.1-**B-C-Top**), the fast rhythms(Fig.1-**B-C-Middle**)and the filtered fast rhythms (Fig.1-**B-C-Bottom**). We found that altering the long-distance coupling significantly affected the system's dynamics. Precisely, we identified periodic (not shown) and chaotic dynamics (Fig.1-**B-C**) of the slow rhythms. The latter were able to generate gamma rhythms with the observed variability in the data, without the need for noise, as in previous modelling studies [2]. This constitutes an alternative mechanism for gamma rhythm generation that, to our knowledge, has not been reported. We then investigated in detail how the dynamics of slow rhythms change with increasing long-range coupling. To quantify such change, we consider the cycle-to-cycle peak values (PV) of the slow rhythms and their relative timing or inter-peak interval (IPI) (see Fig.1-**D**). We plotted these quantities (Fig.1-**E-F**) for the two slow rhythms and found different regimes. For small values, $0 \lesssim c_L \lesssim 3.30$, the system is periodic, and the two slow rhythms exhibit phase synchronization (not shown). For $3.30 < c_L \lesssim 4.11$, the system remains periodic, but the periodicity becomes complex, with several frequencies and cycle-to-cycle peaks emerging for the same value of c_L. For $4.11 < c_L \lesssim 4.96$, the system has an infinite number of frequencies characteristic of chaos (see Fig.2-**E-F**). The transition from the complex periodic regime to the chaos we just described is known as a period-doubling transition to chaos. Moreover, in the chaotic regime, the system exhibits symmetry breaking, where one of the two identical chaotic oscillators has a larger PV and longer IPI compared to the other (see Fig.2-**B-Top**). The identity of the oscillator that acquires the larger PV and the longer IPI depends on the initial conditions. This is known as asymmetric chaos. Further increasing c_L, another period-doubling transition ($4.96 < c_L \lesssim 5.43$) to another chaotic regime ($5.43 < c_L \lesssim 6.91$) occurs. In the new chaotic regime ($5.43 < c_L \lesssim 6.91$), both networks cover a similar range of PV and IPI values (see Fig.2-**E-F**). But the situation is more complex. As the dynamics of the slow rhythms evolve, during an epoch of time, one of the two networks has larger PV and longer IPI than the other; however, in contrast to the former case, this behaviour is reversed and the other network acquires larger PV and longer IPI than the former oscillator, during the next comparable epoch of time (see Fig.2-**C-Top**). The exchange of oscillator identity with the larger PV/longer IPI occurs dynamically. This is known as symmetric chaos. Finally, from $6.91 < c_L \lesssim 8$, the system becomes periodic again. We focus on the two chaotic regimes since they generate gamma rhythms consistent with neural recordings. We considered filtered-fast gamma rhythms (corresponding to fast rhythms filtered in the gamma band (30-100 Hz)). We investigated whether their phases exhibit epochs of stable phase offset and the mechanisms underlying them.

Long-distance communication between the fast brain rhythms.

To quantify the communication between the fast gamma rhythms of each of the two regions and the effect of slow rhythms on such a mechanism, we define the phase locking value (PLV) as

$$\mathrm{PLV} = \frac{1}{N}\left|\sum_{t_x=1}^{N} \exp\left(i(\Delta\phi_k(t_x))\right)\right| \tag{5}$$

where $\Delta\phi_k(t_x) = \phi_k^1(t_x) - \phi_k^2(t_x)$ is the phase-difference between slow or filtered-fast rhythms of the first and second regions at time t_x.

We link communication and synchronization via the communication-through-coherence hypothesis [12]. For the two gamma rhythms to communicate, they must show coherence or phase synchronization. We then asked ourselves the following questions: 1) Do gamma rhythms between the two brain regions show consistent and stable phase-offset relations? 2) If yes, how do slow rhythms shape the phase-offset relations observed between the fast gamma rhythms? To answer both of these questions, we investigate the time series of the slow $E_s^{1,2}$, the filtered-fast rhythms $V_{E_f}^{1,2}$, their phase-differences $\Delta\phi_k$ and their distributions, depending on the values of the parameters c_L and c_{EE} (Fig.2). We found that when the system of coupled slow rhythms exhibits chaotic behavior, the phase offset between them remains remarkably stable (see Fig.2-**A,C**). Therefore, the slow networks exhibit phase synchronization in their chaotic regimes. We found differences between the regimes of asymmetric and symmetric chaos mentioned previously. In the first regime, one slow network always leads the other (Fig.2-**A-Top**); thus, the corresponding phase difference is always positive or negative, depending on which network leads. Chaotic activity serves as input to fast networks, inducing gamma rhythms that resemble in vivo recordings. Interestingly, the generated gamma rhythms exhibit long epochs of stable phase offsets. Unlike the phase difference between

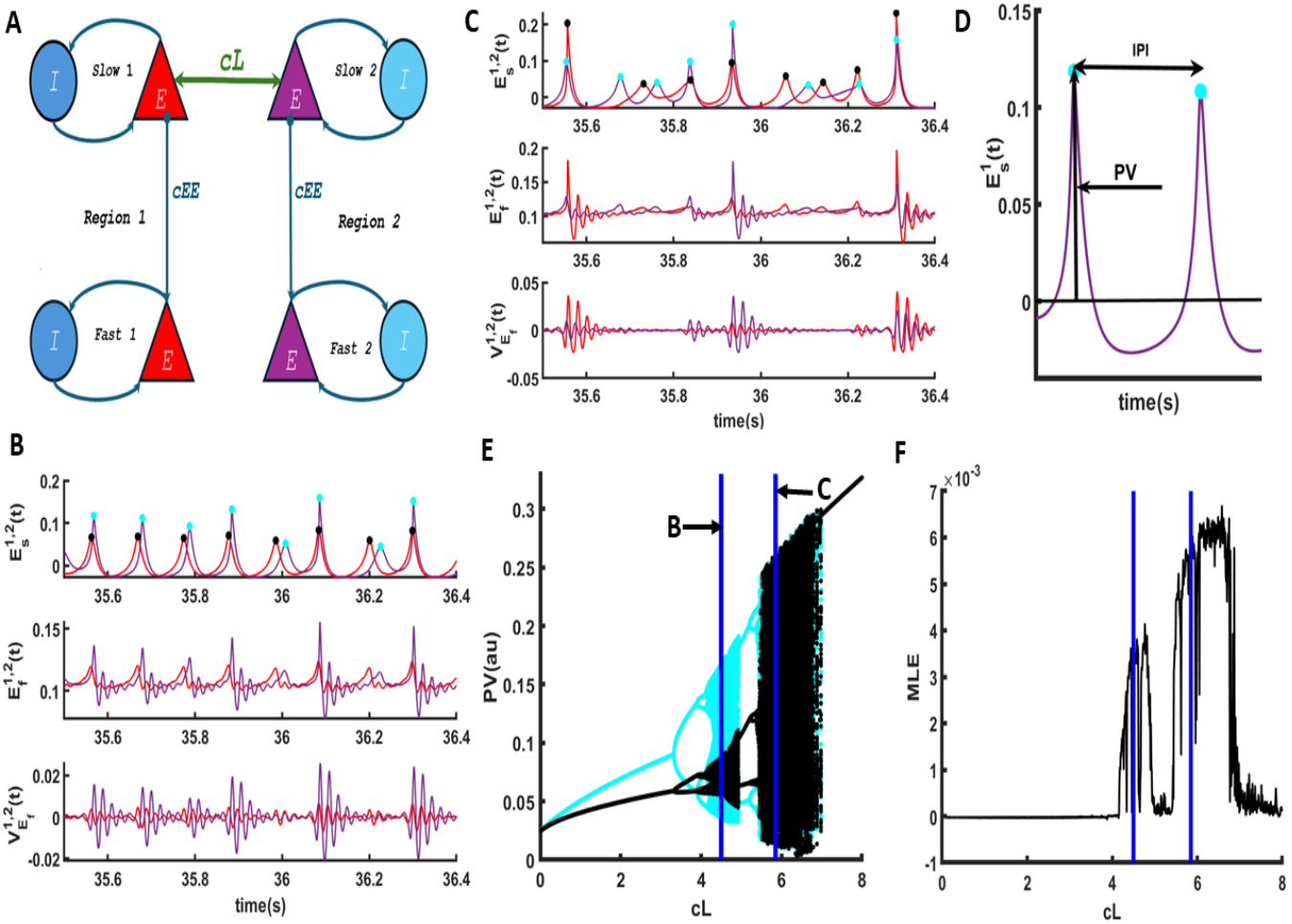

FIG. 1. **Model of long-range coupling of neural rhythms. (A):** Schematic of the coupled neural networks with two identical brain regions connected through a long-range excitatory coupling. A brain region with excitatory (E) and inhibitory (I) neuron populations oscillating at high (gamma) frequency receives input from a network with a similar architecture that oscillates at a much lower frequency. **(B-C):** Time-series of the E activities of the slow (Top panel), fast (Middle panels), and filtered-fast (Bottom panels, where filtered-fast refers to filtering in the 30- 100Hz gamma band), for two values of the long-range coupling c_L. The black and magenta dots show the peak at each oscillating cycle. **(D):** Illustration of the Peak Value (PV) of an oscillation cycle and the Inter-Peak-Interval (IPI) between two consecutive peak oscillations. **(E-F):** Bifurcation diagrams showing the PV and the IPI as the long-range coupling c_L increases, respectively. The values of the long-range couplings used for Timeseries in **(B)** and **(C)** are highlighted with vertical blue lines.

the slow rhythms, the phase difference between the fast rhythms exhibits epochs of positive and negative values. However, there is an asymmetry in the sign of the slow-network phase difference. Precisely, if the phase difference between the slow networks is always positive, the phase difference between fast rhythms will show more epochs of positive stable phase-offsets and vice-versa (see Fig.2-**B**). Despite the absence of structural connectivity between the two distant fast networks, the slow chaotic activities can induce long-range phase synchronization between the gamma rhythms. The nature of the synchronization suggests an asymmetric but bidirectional communication between the gamma rhythms. In addition, the exchange of leadership between the two gamma rhythms is dynamic and entirely controlled by chaos in a nontrivial manner. In the regime of symmetric chaos, both slow networks experience epochs during which one leads the other. The exchange of leadership between the two slow networks occurs dynamically and chaotically. The symmetric chaotic inputs generate realistic gamma rhythms. The phase difference between the two gamma rhythms shows epochs of both positive and negative values consistent with the slow rhythms. Specifically, when a slow network in one region leads, the corresponding fast network also leads during approximately the same epoch, and vice versa. The number of occurrences of epochs of positive and negative phase-offsets is equal in the limit of infinite simulation time. This leads to bidirectional, symmetric long-range communication between the two gamma rhythms. Flexible bidirectional (symmetric and

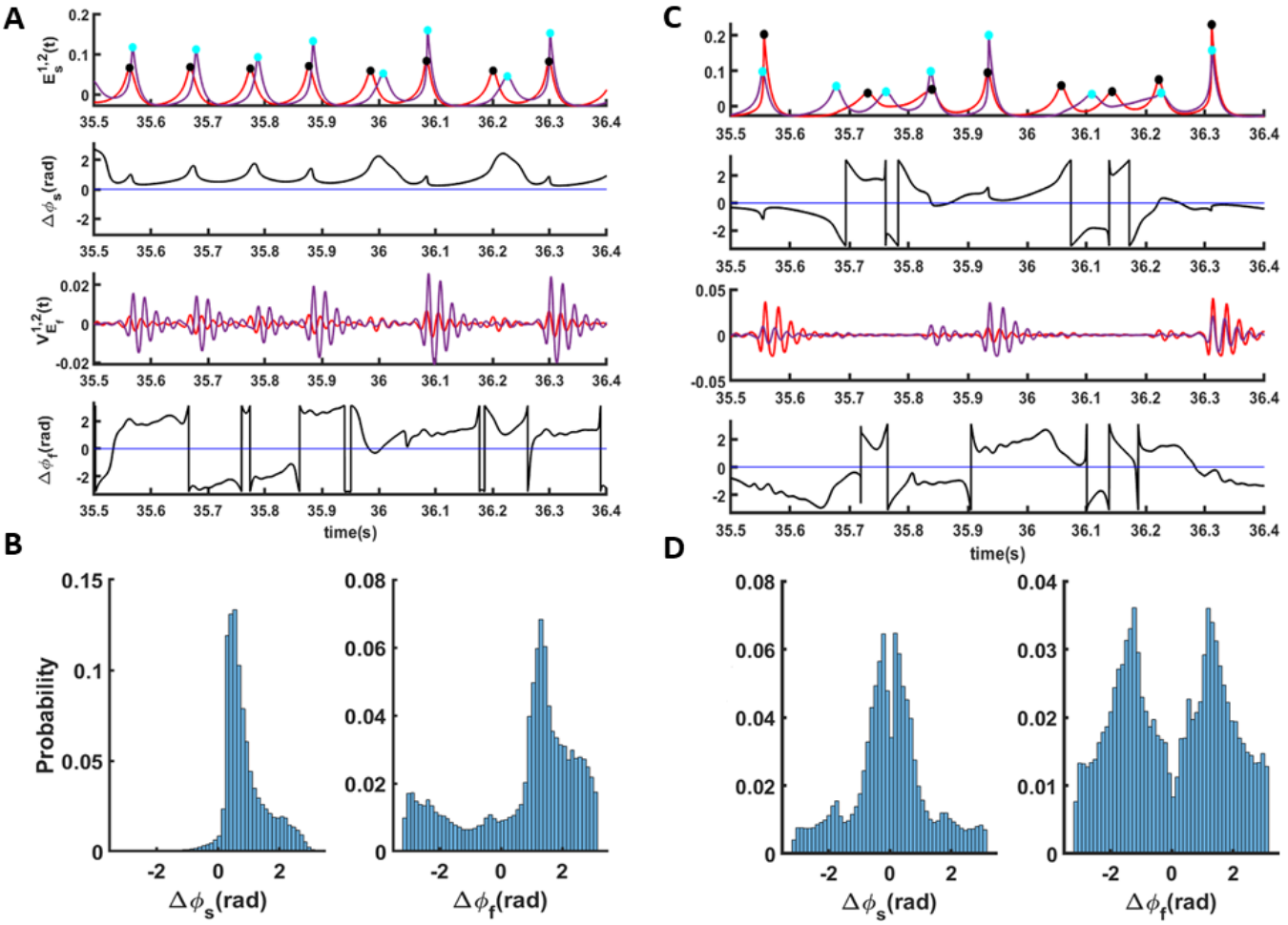

FIG. 2. **Long-range out-of-phase locking induced by chaos. (A):** Timeseries of the slow rhythms (**Top**) their phase-difference (**Middle-Top**) and fast rhythms (**Middle-Bottom**) and their phase-difference (**Bottom**). **(B):**Phase-difference distributions of the slow (**Left**) and fast(**Right**) rhythms. For **(A)** and **(B)**, the parameters are chosen so that the system of coupled networks follows the dynamics shown in Fig. ref Fig1-**(B)**. **(C-D):** Same as in panels **(A)** and **(B)**, but the parameters are chosen so that the system of coupled networks now follows the dynamics shown in Fig.1-**(C).**

asymmetric) communication between two nearby connected gamma rhythms has been reported in a previous study [13]. However, to our knowledge, this is the first report of such flexible communication between distant, unconnected gamma-rhythm networks. Moreover, the former computational work relied on noise, direct connectivity, and propagation delays, whereas our study focuses on a new and utterly different mechanism: chaos. The idea that chaos serves as a generator of gamma rhythms and a coordinator of their timing is consistent with in vivo recordings. Previous work has demonstrated that the synaptic activities of a network of coupled excitatory and inhibitory neurons can exhibit chaotic behaviour in a balanced regime. It thus makes sense for such chaotic inputs to generate random gamma rhythms while also allowing those rhythms to remain consistently locked.

Properties and robustness of long-distance communication To quantify long-distance communication, we analyzed the phase difference time series and distributions between slow and fast rhythms in both asymmetric and symmetric regimes (Fig. 2). We calculated the Phase Locking Value (PLV) and peak locations of these distributions, averaging over 500 initial conditions and repeating the process as the chaotic input strength to the fast rhythms increased.

In the asymmetric chaos regime, with initial conditions IC1 (first slow network leads) and IC2 (reverse), the mean PLV between slow rhythms remained constant (Fig. 3-**A**). Conversely, the mean PLV between fast rhythms decreased with stronger chaotic input (Fig. 3-**B**), indicating degraded long-distance synchronization. The slow rhythm phase difference showed a single peak (positive or negative, depending on IC1/IC2) (Fig. 3-**C**). Intriguingly, for each initial condition, the fast-rhythm phase-difference distribution revealed two asymmetric peaks at positive and negative values (Fig. 2-**B**, Fig. 3-**D**), with differing amplitudes and absolute locations. These locations, however, became increasingly symmet-

ric as the strength of the chaotic input increased. This highlights a unique scenario where networks with distinct inputs exhibit nearly identical dynamics.

In the symmetric chaos regime, both initial conditions yielded identical phase relations between the slow and fast rhythms. The mean PLV between fast rhythms continued to decrease with increasing chaotic input. However, the peaks and locations of the rapid rhythm phase difference distributions remained symmetric as input strength increased (Fig. 4-**B**). This indicates that in the symmetric regime, long-distance phase synchronization is directly governed by the synchronization of the chaotic inputs: a leading slow chaotic network corresponds to a leading fast network, and vice versa.

In both asymmetric and symmetric cases, chaotic inputs from the slow networks can induce gamma rhythms in the fast networks and coordinate their timing. We have illustrated this mechanism for two working points, but it generalizes to all the chaotic regimes.

Cross-frequency phase-phase coupling explains long-distance synchronization. A prominent neural basis for long-distance synchronization is cross-frequency coupling [8], which describes the coupling between neural rhythms across different timescales. However, the original formulation of CFC is restricted to periodic systems. Here, we investigate whether CFC could be generalized to chaotic rhythms. We focus on phase coupling between the fast and slow rhythms in each region. Then, we extend the definition of the PLV in Eq.5 to n cycles of the slow rhythms and m cycles of the fast ones. This leads to the following definition.

$$\text{mnPLV} = \frac{1}{N}\left|\sum_{t_x=1}^{N} \exp\left(i(\Delta\phi_{sf}^{mn}(t_x))\right)\right| \tag{6}$$

where $\Delta\phi_{sf}^{mn}(t_x) = n\phi_s^k(t_x) - m\phi_f^k(t_x)$ is the phase-

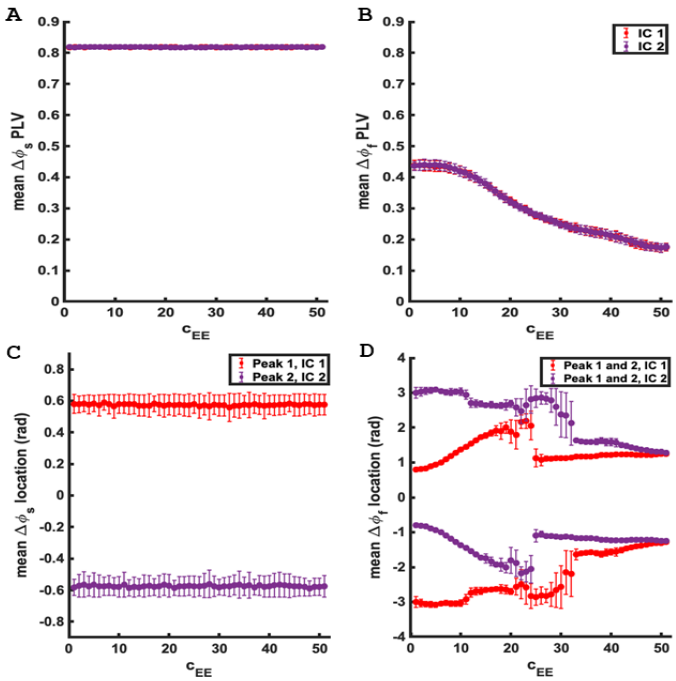

FIG. 3. **Asymmetric and flexible long-range out-of-phase locking. (A):** Mean Phase-locking value between the two slow rhythms as their coupling toward the fast rhythms increases. **(B):** Mean Phase-locking value between the two fast rhythms as the coupling received from the slow rhythms increases. **(C):** Locations of the peak's phase-difference distribution of the slow rhythms for two distinct initial conditions. Initial condition 1 (IC 1) denotes the first slow network leading the second, whereas initial condition 2 (IC 2) denotes the opposite situation. **(D):** Locations of the peak's phase-difference distribution of the fast rhythms corresponding to initial conditions 1 and 2. For a single peak in the phase-difference distribution of the slow rhythms, two peaks are created in the fast rhythms' phase-difference distribution. This suggests a flexible, long-range phase synchronization in which leadership between the two rhythms is dynamically exchanged.

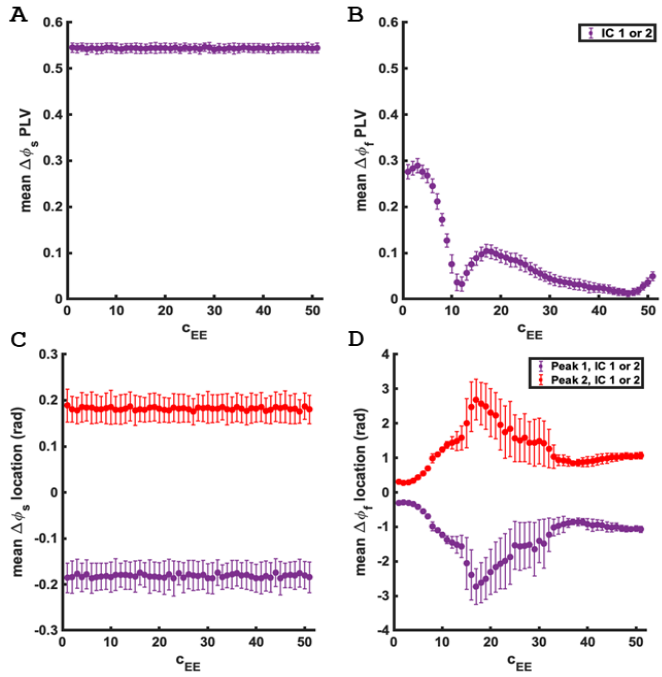

FIG. 4. **Symmetric and flexible long-range out-of-phase locking. (A):** Mean Phase-locking value between the two slow rhythms as their coupling c_{EE} toward the fast rhythms increases. **(B):** Mean Phase-locking value between the two fast rhythms as the coupling c_{EE} received from the slow rhythms increases. **(C):** Locations of the peak's phase-difference distribution of the slow rhythms for any of the two previous initial conditions described in Fig.3. **(D):** Locations of the peak's phase-difference distribution of the fast rhythms corresponding to initial conditions 1 or 2 indistinctly. The phase-difference distribution of the slow rhythms shows two peaks. Those peaks induced two peaks in the phase-difference distribution of fast rhythms. The transition between the peaks occurs dynamically across both slow and fast rhythms.

difference between slow (s) and filtered-fast (f) rhythms of each region ($k = 1, 2$) at time t_x. We consider one cycle of the slow rhythm ($n = 1$) and increase the number m of cycles of the fast rhythms. For the two chaotic regimes (asymmetric and symmetric), we found that the $mnPLV$ shows a dominant peak for $m \approx 6$ (Fig.5). These results suggest a Phase-phase coupling mechanism [14], in which a cycle of the slow rhythm is locked to approximately six cycles of the fast rhythm in each region. We now have a clear picture of the mechanism of long-distance phase synchronization induced by chaos: slow chaotic inputs are phase-synchronized; when they target fast networks, they induce fast rhythms. In addition, they provide a specific phase relationship (via PPC) among these rhythms, thereby enabling their coordination across distant brain areas.

Conclusion We have uncovered a new mechanism by which coordinated slow chaotic inputs induce long-distance synchronization of fast brain rhythms. Our proposed mechanism is consistent with *in vivo* observations and constitutes a realistic mechanism (PPC) by which the brain communicates local information across distant regions. Unlike previous modelling work [11], our approach accounts for the stochastic nature of fast brain rhythms as observed *in vivo*. Our proposed mechanism is robust and constitutes the first modelling work to consistently explain the mechanisms underlying long-range synchronization of fast brain rhythms.

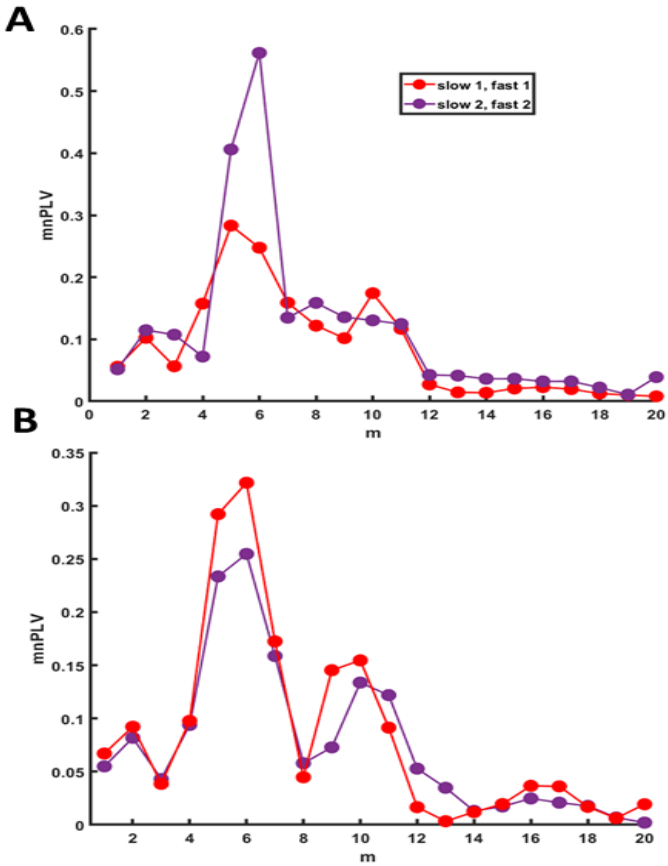

FIG. 5. **A phase-phase coupling mechanism explains the long-range synchronization between fast brain rhythms.** .

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